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Project Title:	The role of senseless in sensory organ precursor selection, and its vertebrate homologue, GFI1, in hair cell differentiation		
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Task Description:	Hair cells function as mechanoreceptors to transduce gravity, balance, and hearing. We have previously shown that the generation of hair cells is dependent on the expression of the Mouse atonal homologue 1 (MATH1)in precursors of hair cells. Most recently, injection of adenovirusus encoding Math1 into the inner ear fluid of adult animals has been shown to regenerate hair cells that are innervated. Unfortunately, regeneration is not very efficient, similar to what is observed when fly atonal alone is overexpressed in flies. However, coexpression of atonal and senseless, another transcription factor, is extremely efficient in specifying numerous extra sensory organs in flies. As we have previously shown that senseless, similar to atonal, is necessary and sufficient to form sensory organs in flies, we propose to study the role of senseless in much more detail in specific aim 1. In addition, we also propose to study the role of the vertebrate senseless homologue, Gfi1 in specific aim 2, as we have shown that this gene is also required for proper hair cell differentiation and function. We therefore wish to further our understanding of the biology of this gene in vivo, and test if the		

	synergism observed between atonal and senseless is evolutionary conserved. We hope that this research will allow us eventually to regenarte hair cells in adult animals that have lost functional hair cells. To understand how hair cells which are required for hearing and proprioception develop. To use this information to plan efficient strategies to regenerate hearing and propioception in humans that have lost these cells. The significance of our research has been highlighted recently with a publication in Nature Medicine showing that our premises that Math1 can regenerate hearing in deaf mammals is correct.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have made numerous important discoveries in the past year. With respect to specific aim1 we are in the process of submitting two manuscripts. In the first manuscript we present convincing evidence that the main function of the proneural genes, including scute, achaete, amos, and atonal is to activate senseless. In the absence of these proneural genes but in the presence of Senseless protein, we are able to produce external sensory organs in wing and thorax. However, the reverse is not true, i.e. in the absence of senseless, ectopic expression of proneual genes cannot induce external sensory organ formation. In addition, another gene, named daughterless, which is ubiquitously expressed and is known to heterodimerize with proneural proteins, is instrumental for sensory organs specification. Hence, daughterless together with senseless are necessary and sufficient for the specification of sensory organs. This is especially important for the wing margin stout bristles where loss of proneural proteins has been known for more than 20 years not to cause a loss of the these bristles. It turns out that the precursors of these bristles can be specified by senseless and daughterless alone. Unlike in other tissues, early expression of senseless in the wing margin is not dependent on the expression of proneural proteins, but rather on the presence of Wingless. We have shown that highly elevated levels of Wingless activate senseless transcription, and since daughterless is ubiquitously expressed, both proteins are sufficient to specify the wing margin bristles in the absence of proneurals. Moreover, the proneural genes are required for the survival of the cells after they have been specified. Hence, the proneual genes in this context function as survival factors. This loss of cell survival is probably due to the fact that the senseless levels are not maintained properly at a later stage during organ differentiation in the absence of proneural proteins. This defect can be overcome by ectopic expression of the anti-apoptotic protein P35, which leads to the survival of all sensory neurons and accessory cells in the wing margin. Surprisingly, upon inhibition of apoptosis the non-sensory bristles of the posterior wing margin acquire neuronal fate and send out axons to the CNS. Since proneural proteins are not expressed in the precursors of the posterior margin bristles, it has long been thought that they don't have the potential to become sensory organs. We find, however, that presence of senseless and daughterless is enough to endow neuronal potential to these cells, although they normally undergo apoptosis, presumably because of a lack of proneural gene expression. In summary, our data show that senseless is the proneural protein for the wing margin bristles and that proneural genes function as survival factors, not in sensory organ specification. This also explains why there is such a strong synergism between senseless and proneural genes in other sensory organs. This work will soon be submitted for publication. In addition to the above observations, we have initiated a structure-function analysis of the Senseless protein. Our data indicate that the most important domain of the protein is indeed the most evolutionarily conserved domain: the four zinc fingers. We find that zinc fingers have two separable functions: DNA binding, and protein-protein interactions. Our data show that DNA binding of Senseless is responsible for transcriptional repression, whereas binding of Senseless to proneural proteins is at base of the synergistic interaction between proneural proteins and Senseless. We have accumulated biochemical protein interaction data, DNA binding data, transcriptional assay data, and in vivo data to support the following model. With exception of the wing margin bristles, proneural proteins activate senseless transcription. At low levels, Senseless initially mildly represses transcription of the proneural proteins. However, this repression is turned into a very potent transcriptional activation when the level of Senseless reach a certain threshold. Senseless then binds the proneural proteins, obscuring the transcriptional repression, and functioning as a transcriptional activator of proneural genes. This work will soon be submitted for publication. Most recently we have shown that a vertebrate homolog of Senseless, Gfi1b is able to produce numerous extra bristles in the fruitfly when ectopically expressed. We surmise that the synergism between vertebrate proneural proteins and Gfi1 proteins will be evolutionarily conserved. We are currently testing this hypothesis using in vivo coexpression of vertebrate proneural proteins and Gfi1B as well as through transcription assays. This may have important implications in the regeneration of hair cells in vertebrates. A report in Nature Medicine has recently shown that ectopic expression of MATH1 is sufficient to not only regenerate hair cells in deaf animals, but that it also partially restores hearing, an observation that we had previously predicted on the basis of our fly data. Hence, coexpression of Gfi1 and MATH1 may synergize and very efficiently regenerate hair cells in cochlea and balance organs, similar to what we observe in fruitflies. In an another interesting twist we recently found, in collaboration with Dr. Huda Zoghbi's laboratory, that Gfi1 is one of the main mediators of the degeneration of the cerebellar Purkinje Cells (PCs) in Spinocerebellar Ataxia type 1 (SCA1). Senseless and Gfi1 bind the SCA1 protein in vitro and in vivo. Overexpression of the SCA1 protein results in loss of Gfi1 protein in PCs and also loss of Senseless in sensory precursors in flies. Gfi1 is highly expressed in the PCs and Gfi1 mutant mice exhibit cerebellar degeneration. Furthermore, we find dosage-dependent genetic interactions between Senseless/Gfi1 and SCA1 in both mice and flies. Our data strongly suggest that abundant SCA1 protein slates the Gfi1 protein for degradation in the proteasome. To our knowledge, this is the first study to offer a molecular mechanism for the selective nature of neuronal degeneration in a neurodegenerative disease despite the ubiquitous expression of the mutant SCA1 protein. This work was recently submitted to Cell and is in second revision.
Bibliography Type:	Description: (Last Updated: 09/03/2009)
Articles in Peer-reviewed Journals	Jafar-Nejad H, Bellen HJ. "Gfi/Pag-3/senseless zinc finger proteins: a unifying theme?" Molecular and Cellular Biology. 2004 Oct;24(20):8803-12. Review. PubMed PMID: 15456856 , Oct-2004
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